THE ASSOCIATION OF MATHEMATICS TEACHERS OF INDIA

AMTI – NMTC – **25th October, 2025** – PRIMARY FINAL

Timings: 10am to 1pm

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Instructions:

1. There are 8 questions, answer all questions. Each question carries 10 marks.
2. Elegant and innovative solutions will get extra marks.
3. Diagram and justification should be given wherever necessary.
4. Before start answering, fill in the FACE SLIP completely.
5. Your **Rough Work** should be in the answer sheet itself.

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1. ABC is a triangle in which ∠CAB:∠ABC: ∠BCA = 11:4:3. The line through point C making an angle $\frac{\angle BCA}{3}$ with BC meets the line through point B which bisects ∠ABC, at D. The bisector of ∠CAB and the line through D making an angle $\frac{\angle BDC}{3}$ with CD, meet at E. Find and write the measure of ∠AED.

2. Consider the seven-digit number $5a793a4$, where a is a digit. Find all such seven-digit numbers which are divisible by 3. Write down the smallest and the largest of these numbers and write the difference between these two numbers.

3. N is a two-digit number. The digit 3 is affixed to the right of N to make it a three-digit number. The new number is 777 more than the original number. Find N.
   M is a two-digit number. M is equal to 4 times the sum of its digits. Find all such two-digit numbers. Let P be the average of such all two-digit numbers. Calculate $N + P$.

4. Look at the sequence 1, 1, 2, 3, 5, 8, 13, … . This is called the Fibonacci sequence. Starting from the third term, each term is the sum of its immediate previous two terms.
   For Ex: 5 = 2 + 3, 8 = 3 + 5, 13 = 5 + 8 etc.
   Find the remainder when the number in the 2025^{th} term is divided by 3.

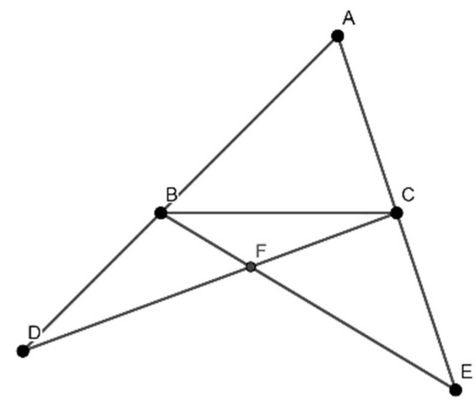
5. In an igloo pack there are two flavours of cup ice-cream, vanilla and butterscotch. First, Pramod removes 36 vanilla ice cups and 4 butterscotch cups. The remaining cups of ice-cream are divided into 11 groups. In each group, there are 8 vanilla and 3 butterscotch ice-cream cups. What is the greatest number of ice-cream cups in the pack if at least 80% of the total ice-cream cups are vanilla ice-cream cups?

6. After the complete simplification of the fraction

$$\frac{\left(\frac{1}{3}+\frac{1}{4}+\frac{1}{5}+\frac{1}{6}\right)}{\left(\frac{1}{3}-\frac{1}{4}+\frac{1}{5}-\frac{1}{6}\right)}+\frac{\left(\frac{1}{3}+\frac{1}{4}-\frac{1}{5}-\frac{1}{6}\right)}{\left(\frac{1}{3}-\frac{1}{4}-\frac{1}{5}+\frac{1}{6}\right)},$$

The result is of the form $\frac{a}{b}$, where a and b are natural numbers with no common factor other than 1.
What fraction to be subtracted from $\frac{a}{b}$ to get 1, what is the integer part of this fraction?

7. In triangle ABC, point D is taken on the extended side AB and point E is taken on the extended side AC such that BC = BD = CE. Line-segments BE and CD are drawn, which intersect each other in F. The measure of angle BAC = 70º, then find the measure of angle BFD. Justify your answer.



8. Using the digits 0, 2, 4, 6, 8 each at least once form and write the greatest and the smallest seven-digit number divisible by 99. Justify your answer.
Also, find and write the difference between these numbers.

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